Sushant Kamble

Human–Computer Interaction
Extended Reality
Wearable Interfaces

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Austin, TX

RESEARCH PROFILE

- Engineer and aspiring XR researcher with an M.S. in Information Studies from UT Austin and over six years of industry experience in software engineering and UX design.
- Research interests span embodied interaction, wearable interfaces, and tactile feedback systems that bridge physical and virtual spaces.
- Experienced in building complex interactive systems, rapid prototyping, hardware integration, and user-centered evaluation to advance experimental research in immersive environments.

EDUCATION

The University of Texas at Austin **Austin, TX**
Master of Science in Information Studies *Aug 2022 – May 2024*

- Focus: human–computer interaction, immersive technologies, and user-centered evaluation.

Savitribai Phule Pune University **Pune, India**
Bachelor of Engineering in Information Technology *Jul 2015 – Jun 2018*

- Focus: mobile and web application development, embedded systems.

RESEARCH AND PROTOTYPING PROJECTS

Phantom Touch | Wearable Haptics for Virtual Surface Interaction
Independent Research Project (In Progress) *2026 – Present*

- Investigating wearable haptics for virtual surface perception, placing actuators on the backs of fingers to preserve direct physical interaction.
- Developing custom middleware to capture Meta Quest hand-tracking and Unity touch-event data and relay haptic commands to ESP32-S3-controlled wearable actuators.
- Planning to investigate movement-dependent texture rendering using UV and roughness maps to generate high-fidelity vibration patterns for linear resonant actuators.
- Planning perceptual evaluations to examine how texture detail, finger movement, hand tracking uncertainty, and actuator response affect texture discrimination.

Elixir Expedition | Attention-Dependent Environmental Changes in VR
Academic XR Research Project | UT Austin | Four-Person Team *2023*

- Co-developed a Unity/Meta Quest 3 experience exploring change blindness, perceptual awareness, and embodied interaction.
- Led experimental design, VR prototyping, and standalone performance optimization; implemented environmental changes triggered by head orientation and proximity.
- Conducted an exploratory evaluation with eight participants, gathering feedback on awareness and recall of environmental changes.

Solar-Assisted Smartphone Charging Case | Portable Hardware Design
Undergraduate Hardware and Entrepreneurship Project | Two-Person Team *2018*

- Co-developed a portable charging prototype integrating solar energy harvesting, supercapacitor storage, battery management system, and a wireless Android companion app.
- Iterated circuit and enclosure designs to balance power efficiency, portability, and form factor—constraints relevant to wearable XR hardware.
- Registered the venture through the university's Technology Business Incubation Center and received mentorship and research funding under the Make in India initiative.

RESEARCH EXPERIENCE

The University of Texas at Austin **Austin, TX**
Graduate Research Assistant (Advisor: Shefali V. Patil) *2023 – 2024*

- Developed automated AI-enabled data-acquisition pipelines to collect and process unstructured data from Wikipedia, IMDb, and the GDELT dataset to support research.
- Developed customized Qualtrics survey interfaces using Qualtrics APIs, implementing conditional branching, eligibility screening, and automated disqualification logic for large-scale participant studies.

INDUSTRY EXPERIENCE

Beheld Health **Austin, TX**
Frontend / User Experience Engineer *Jul 2024 – Present*

- Lead frontend development of a production React Native healthcare application, translating outreach, scheduling, and task-management workflows into usable interfaces.
- Build reusable iOS/Android components, integrate Twilio communication, and oversee frontend architecture, performance, and reliability.
- Collaborate with researchers and healthcare professionals; bring experience relevant to participant-facing applications, wearable companion apps, and maintainable experimental software.

Scalex Cloud **Pune, India**
Software Engineer II *Jul 2018 – Jul 2022*

Client: Clarista Inc

- Led frontend development and team delivery for Clarista's visual data platform.
- Built Flows, a no-code editor using React, Redux, and D3.js, enabling drag-and-drop data transformation.
- Developed data cataloging, lineage visualization, and a real-time analytics dashboard builder.
- Established reusable components and a shared design system to improve consistency and development efficiency.

Client: Microsoft Research

- Developed a dynamic React survey deployed through Amazon Mechanical Turk to collect text-classification data for Cortana model training, gaining experience building remote data-collection interfaces.

TALKS AND WORKSHOPS

Blending Realities | XR Workshop **Austin, TX**
Bezi Fellow | Co-organizer and Facilitator | UT Austin *Feb 2024*

- Co-organized a beginner-friendly XR workshop with a five-person team using browser-based Bezi to reduce setup barriers.
- Developed hands-on demonstrations of 3D interaction, WebXR, and texture optimization
- Supported novice prototypers and communicated XR concepts for lab onboarding, teaching, and outreach.

Unity Optimization for Meta Quest **Austin, TX**
Technical Talk | UT Austin *Nov 2023*

- Delivered practical instruction on standalone VR rendering, including occlusion culling, batching, LOD, baked lighting, and dynamic lightmap switching.
- Demonstrated performance profiling with Meta Quest Developer Hub and OVR Metrics Tool, supporting stable immersive prototypes for interaction studies.

SKILLS

XR Development	Unity, XR Interaction Toolkit, WebXR, Three.js, Meta Interaction SDK
3D & Spatial Design	Blender, Figma, Spatial UI Design (Bezi), VR Performance Optimization
Embedded Systems & Physical Computing	Embedded Programming, Sensor Integration, Interactive Hardware, Digital Fabrication, Physical Prototyping
Programming & Frontend	JavaScript, TypeScript, Python, C#, React, React Native, Redux, Next.js, HTML, CSS
Backend & Infrastructure	Node.js, Express.js, REST APIs, AWS Lambda, Firebase, MongoDB, MySQL, Git
Research & Evaluation	Experimental Prototyping, User Studies, Qualtrics, Heuristic Evaluation, Interaction Design, Data Visualization, Automated Data Collection
Development, Testing & Monitoring	CI/CD, Build and Deployment Automation, Error Monitoring, Browser Automation, XR Performance Profiling